



CSCR2600 - Project-Based Learning (PBL) -1

JAYMANI KRISHI SEWA — Digital Agritech Marketplace for Farmers & Local Vendors

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Project Title

JAYMANI KRISHI SEWA — Digital Agritech Marketplace for Farmers & Local Vendors

Team / Group Formation

S. No	Student Name	Roll Number	System ID	Role
1	Manish Pandey	2515010348	2025266283	Tester & Developer
2	Jayant Raj	2515010273	2025298708	Designer & Developer

Technologies to be used

Software Platform & Backend:-

- Framework: Python (Flask)
- ORM / Data Access Layer: SQLAlchemy
- Database: MySQL
- Supporting Stack: Redis, Docker, Nginx, Gunicorn
- APIs: Razorpay

Front-end:-

- Core Languages: HTML5, CSS3, JavaScript (ES6+)
- Interfaces: Responsive farmer, vendor, and admin workflows.

Hardware Platform:-

- Processor: Intel Core i5 / AMD Ryzen 5
- RAM: 8 GB or higher
- Storage: Minimum 256 GB SSD
- OS: Windows, macOS, or Linux (Ubuntu)
- Editor/IDE: Visual Studio Code, PyCharm
- Web Browser: Google Chrome, Microsoft Edge

Tools

1. Docker (Vendor: Docker, Inc.) - Containerized deployment
2. Redis (Vendor: Redis Ltd.) - In-memory caching
3. Razorpay API (Vendor: Razorpay) - Secure payment gateway
4. Nginx (Vendor: F5, Inc.) - Reverse proxy
5. Gunicorn (Open Source) - WSGI application server
6. SQLAlchemy (Open Source) - Object-Relational Mapper (ORM)

Problem Statement

The agricultural retail sector heavily relies on traditional, manual workflows. Currently, there is no simple, hyper-local platform that seamlessly connects farmers with verified local agricultural retailers while combining product discovery, inventory visibility, and digital credit.

Our project specifically targets the severe operational frictions present in daily agri-retail, which include:

- **Price & Stock Uncertainty:** Farmers travel to multiple physical shops without prior knowledge of product availability or local pricing.
- **Fragile Paper "Bahi-Khata" Systems:** Manual credit records create an unreliable financial environment that is difficult to maintain and preserve against disputes.
- **Time & Fuel Loss:** Unplanned trips to distant agri-retailers lead to a substantial loss of time and fuel, hurting farmer productivity.
- **Vendor Digital Limitations:** Local agricultural retailers currently lack accessible digital tools for real-time inventory management.

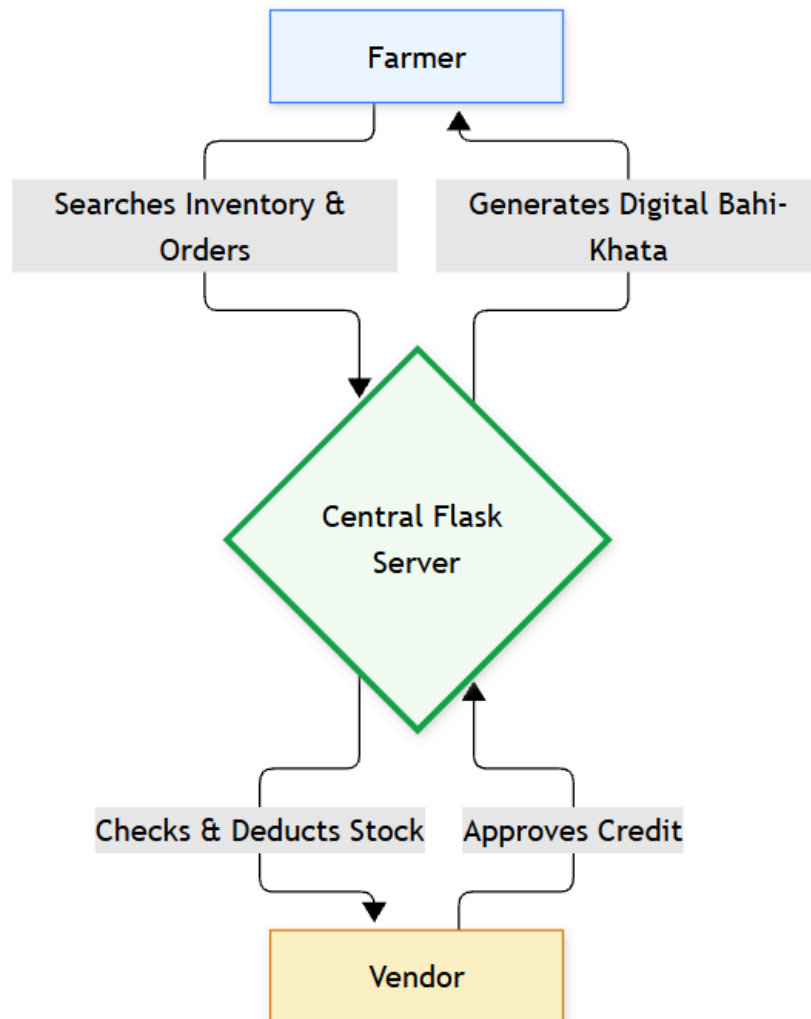
Literature Survey

S.No.	Author(s) & Year	Title	Methodology	Key Findings
1	Sharma, R., & Kumar, V. (2023)	Digitization of Rural Retail	Empirical Analysis	Digital ledgers reduce disputes by 40%.
2	Patel, A., et al. (2022)	Hyper-Local Marketplace Models	Case Study	Farmers lose time due to stock uncertainty.
3	Singh, M. (2021)	Friction in Agri-Retail	Data Mining	Lack of real-time pricing hurts margins.
4	Gupta, N., & Joshi, P. (2020)	Adoption of Mobile Tech	Quantitative	UI/UX is the biggest adoption factor.

Project Description

Jaymani Krishi Sewa is a comprehensive, digital agritech marketplace that bridges the gap between farmers and local agricultural vendors. The primary scope of this project is to integrate hyper-local product discovery, real-time inventory synchronization, and secure digital credit management (Bahi-Khata) into a single, unified web application.

Context Flow Diagram



Project Modules: Design/Algorithm

The system is logically partitioned into five interdependent modules:

- Authentication & Role-Based Access Module: A unified gateway that manages secure logins for all users while strictly isolating data based on user types.
- Hyper-Local Discovery & Search Module (Farmer): The core search engine that allows farmers to find nearby agricultural supplies dynamically.
- Vendor Inventory & Catalog Module (Supply): A robust dashboard giving retailers control over their digital storefront and stock levels.
- Digital Order & Cart Processing Module: Handles the transactional phase where demand meets supply seamlessly without stock conflicts.
- Digital "Bahi-Khata" (Credit Ledger) Module: Digitizes the traditional, paper-based trust system between farmers and local retailers into a secure financial ledger.

Implementation Methodology

The system is being developed using an Iterative and Phased approach. This methodology ensures continuous refinement across our 8 core milestones: Requirement Analysis, UI/UX Design, Backend/Database setup, Frontend Development, System Integration, Testing, and Deployment.

Technology Stack & Tools

Category	Technology / Tool	Purpose
Backend Framework	Python (Flask)	API Routing & Server Logic
Database & ORM	MySQL & SQLAlchemy	Relational data storage and secure transactions
Frontend	HTML5, CSS3, JavaScript	Responsive Farmer/Vendor User Interfaces
Caching	Redis	High-speed in-memory query caching
Deployment	Docker, Nginx, Gunicorn	Containerization and production server routing

Result & Conclusion

Jaymani Krishi Sewa successfully conceptualizes a highly practical, modern solution to the operational frictions inherent in traditional agricultural retail. The main achievement of this project lies in creating a unified ecosystem that simultaneously connects local farmer demand with verified local vendor supply while eliminating critical pain points like blind shop visits and time loss.

The most significant innovation of this project is the modernization of the traditional rural credit system by digitizing the "Bahi-Khata" (credit ledger). In conclusion, Jaymani Krishi Sewa stands out by empowering existing micro-local economies and optimizing supply chain efficiency.

Future Scope and further enhancement of the Project

While the current scope delivers a robust foundation, the platform possesses immense potential for scalable future enhancements:

1. Multi-Lingual & Voice Support: Native regional language support and AI-driven voice search capabilities.
2. AI-Driven Predictive Analytics: Machine learning algorithms to predict crop requirements.
3. Last-Mile Logistics Integration: Integrating local delivery partnerships for door-to-door delivery.
4. Integration with Government Subsidy Schemes: Linking the digital ledger with state/central agricultural subsidy APIs.
5. Native Mobile Application & WhatsApp Bots: Transitioning into native Android/iOS apps with automated chatbot integration.

Advantages of this Project

1. Advantages for Farmers (Primary Beneficiaries): Elimination of blind travel, financial transparency through digital Bahi-Khata, and real-time price comparison.
2. Advantages for Local Vendors (Secondary Beneficiaries): Digital storefront reach, automated ledger management, and inventory optimization.
3. Overall Systemic Advantage: The unification of the local agricultural economy into one streamlined, digital marketplace.

Outcome

The team aims to achieve the following highly scalable outcomes upon project completion:

- Agri-Startup Deployment (Project to Product): Transitioning from an academic prototype to a live pilot program in collaboration with local Mandis and Kisan Vikas Kendras (KVKs), providing real farmers with a functional digital marketplace.
- Agri-tech Innovation & Hackathons: Showcasing the platform under the 'Agriculture & Rural Development' themes at national competitions (e.g., Agri-Hackathon, Smart India Hackathon) to secure funding from Agri-business incubators.

- Agri-Economics Research Publication: Compiling case studies on how digital "Bahi-Khatas" improve farmer livelihoods and reduce supply chain friction, to be published in Scopus Indexed journals focusing on Agricultural Economics and Rural Technology.

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